Exercise 1 – Preprocessing

Please submit a Word document in pairs *by Moodle no later than May 11, 2026*

Make sure to include your IDs in the document

Any questions? Send an email to yoavmelamed@mail.tau.ac.il

Dataset

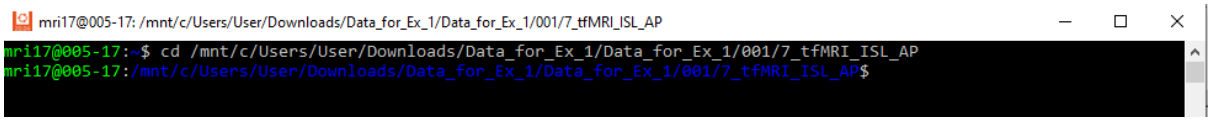
In Moodle you will find the dataset for this exercise. It includes different modalities (T1W and fMRI) of MRI scans for three participants in DICOM format. Each participant's folder contains a different modality. In later stages we will provide more information regarding the fMRI task.

Download the dataset to your computer. Extract the data once downloaded.

If you are using the computer labs' computers - In the following steps you'll see some commands that are meant to be used in the Ubuntu terminal. To navigate to a specific path within Ubuntu, you'll need to used the following command:

cd /mnt/c/<the rest of your path>

Make sure to use the correct slash in your code (use this / and not this \), and make sure you write /mnt/c at the beginning instead of the full path for your directory. For example, this is how you can navigate to the fMRI scan of your first participant, if you've saved the data for this exercise in the download folder (You can see your location in blue):

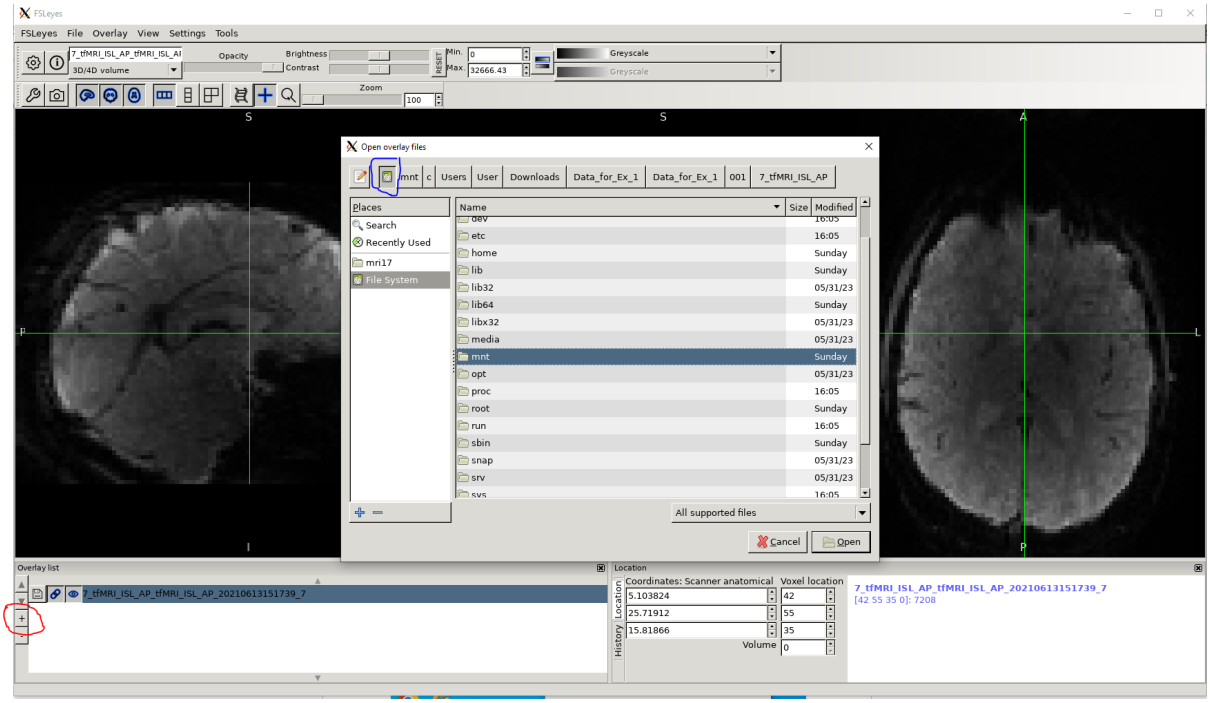


1. Dcm2nii:
   - To convert your dicom files to nii format, you'll need to open the directory in which MRIcroGL is installed in your computers and locate the dcm2niix.exe file (In the computer lab, it's in - C:\MRI_Software\MRIcroGL\Resources). Find the exe named dcm2nii. Open your data directory in another window and drag the folder containing dicom images to the dcm2nii exe (do not double click or open the dcm2nii exe, just drag the folder to it) If you see a warning window - select 'run anyway'. You may see a pop-up window that will warn you about this process, but in the computer lab you can ignore it and press 'run anyway'. The windows command line window (CMD) will open, and you will see the script working.
   - Look in the dicom folder and find the .nii file there.
   - Important: **Do not drag more than one folder** (T1w and fMRI separately, for each participant) at a time. It will corrupt the final .nii file.
   - Open FSLeyes and view your NIfTI files.
     - To open FSLeyes, open your terminal and type: *fsleyes &*

How to open nifti data in fsleyes when working in the computer lab?

When accessing files in fsleyes, you'll need to navigate to the windows storage. For example, if you've downloaded to exercise data to the download folder, this is how you open the files in fsleyes: Click on the + sign inside fsleyes (red circle) and navigate to the home folder of ubuntu (blue circle). Once there, follow the path in the image (/mnt/c/Users….) and find the correct .nii file.

You can also drag the .nii file directly into the fsleyes viewer.



- In FSLeyes, open your Nifti data and look at it. Check the dimension and resolution of your data.

  Question1: What are the dimensions and resolution of all the modalities of participant number 1?

2. Motion correction

   - For ease of use, you can rename the .nii images at this point to shorter filenames - it will make the next parts easier.

   In order to perform analysis on multiple MRI scans, we would like to perform motion correction (register each 3D volume in a 4D data set to a specific single 3D volume), registration between different modalities (such as structural scan and fMRI scan) and normalize all participants to the same space.

First, we should perform motion correction for each 4D data set. In later exercises you will perform this using the FEAT GUI, but here we will use the *mcflirt* command line tool.

For each fMRI image type this command in your terminal (make sure you're in the correct directory or provide full path for your file_name):

- *mcflirt -in <your_in_file_name> -plots*

This command will perform rigid body transformation (6 DOF) of your fMRI image to the middle volume. Output will appear in your directory as *<your_in_file_name>_mcf*. The optional *-plots* flag will save the transformation parameters as *<your_in_file_name>_mcf.par* (It's not mandatory, but we recommend saving the motion parameter for further analysis).

Question 2: a. Plot the motion parameters for participant number 2 and add the plot to your report. What is the difference between the first 3 motion parameters and the last 3 motion parameters?

- Try aligning the volumes to the first volume instead of to the middle volume (mcflirt default). *mcflirt -in <your_in_file_name> -refvol 1 -plots -o <new_output_name>*

  (make sure to use a new output name to avoid deleting the previous results)

Question 2: b. Plot the new motion parameters for participant number 2 and add the plot to your report.

3. BET

Now that we have corrected the scans for motion, we can extract the brain from our image and remove the non-brain background.

- Open the FSL GUI and choose BET brain extraction.
- Choose your input image (for 4D data, make sure that you choose the *_mcf* data files) and your output name (we recommend using the FSL default which is *<your_input_image>_brain*).
- In the advanced options, mark the option "Output binary brain mask".
- Now you can run standard brain extraction using bet2 by pressing Go. You can try changing the fractional intensity threshold as explained in the class.
- Perform this for all T1w and fMRI images.
- Optional – you can perform this part using the command line tool with the BET command. Type this command in your terminal (make sure you are in the correct directory):
  - *bet <your_input_image> <your_output_name> -m -f <your_fractional_intensity_threshold>*
  - You don't have to insert a fractional intensity threshold - the default threshold will be 0.5.

Question 3: What, in your opinion, is the best fractional intensity threshold (use FSLeyes to review your data) for the T1w of participant number 3? Add the pre-BET and post-BET brain images to your report.

4. Registration and normalization

In this part we will want to register the fMRI data to the T1w image and then normalize each participant's images to the MNI template. In this exercise we will perform it by using the command line tool, although it can be done using the FEAT GUI.

First, we would like to perform linear registration between the structural image (T1w) and the functional image (fMRI).

- For each fMRI image type this command in your terminal (make sure you're in the correct directory or provide full path for your file_name):
  - *flirt -in <your_input_fMRI_data_after_BET> -ref <your_T1w_image_after_BET> -omat <your_directory>/functional2struct.mat*
  - For the reference image – choose the T1w image **post**-BET.
  - The output (the *-omat* option) of this command is an affine matrix which we will use in later stages. Make sure you save the matrix for a specific participant in the participant folder.

Next, we will perform linear and non-linear registration, from the participant T1w image to a template T1w image in MNI space.

- For each fMRI image type the following command in your terminal (make sure you're in the correct directory or provide full path for your file_name):
  - Run *flirt* on the **post**-BET images:
  - *flirt -in <your_T1w_image_after_BET> -ref <MNI_T1w_brain_image> -omat <your_directory>/struct2standard.mat*
    - The MNI T1w brain image can be found at FSL standard directory: */home/<your_user>/fsl/data/standard/MNI152_T1_2mm_brain.nii.gz*
    - your_user depends on the computer you're using. In the computer lab you can see the user in the Ubuntu terminal). The full directory should be: \\wsl.localhost\Ubuntu-20.04\home\<your_user>\fsl\data\standard
    - Here again the output (the *-omat* option) of this command is an affine matrix which we will use in later stages. Make sure you save the matrix for a specific participant in the participant folder.
  - Run *fnirt* on the **pre**-BET images (keep in mind, this part could take **a long time** for each participant):
  - *fnirt --in=<your_T1w_image_before_BET> --ref=<MNI_T1w_image> --aff=struct2standard.mat --cout=<your_directory>/warp_struct2standard*

- The MNI T1w image can be found at FSL standard directory: */home/<your_user>/fsl/data/standard/MNI152_T1_2mm.nii.gz*
- The input --aff is the previous step's output (the file created from --cout option). Make sure you take the correct file for each participant.
- The output (the *--cout* option) of this command is a warp field which we will use in later stages. Make sure you save the file for a specific participant in the participant folder.
- FNIRT takes a **long time**. In the computer lab it could be up to 20-30 minutes.

Now, we can apply all transformations to our original image and receive the fMRI image in standard space (in our case, MNI).

- For each fMRI image type the following command in your terminal (make sure you're in the correct directory or provide full path for your file_name):
  - *Applywarp -i <your_input_fMRI_data_after_BET> -r <MNI_T1w_brain_image> -o <your_output_name_for_standard_fMRI_data> --premat=<your_directory>/functional2struct.mat --warp=<your_directory>/warp_struct2standard.nii.gz*
  - The inputs for --premat and --warp are the files created in the first flirt (from functional to T1w) and the fnirt. Make sure you take the correct file for each participant.

Make sure to use the correct syntax for each part.

Question 4: Add the final image of participants 1 and 3 to your report. Are they aligned in your opinion? If not, try and place them together in fsleyes and mark any areas that don't look aligned to you. Use a different color for each brain map so differences are evident.

5. Smoothing

For the last part, we would like to perform smoothing on our fMRI data for later statistical analysis. We will perform the FSL standard smoothing of 5mm FWHM.

- For each fMRI image type the following command in your terminal (make sure you're in the correct directory or provide full path for your file_name):
  - *fslmaths <your_input_fMRI_data_in_standard_space> -kernel gauss 2.1233 -fmean <your_output_name>*
  - The sigma value 2.1233 is calculated for a 5mm FWHM smoothing.

Question 5: Add an image of the smoothed fMRI data of participant number 2. What value of sigma would you use if you wanted a 3mm smoothing? Or 7mm?